

All it Takes is a Slight Rotation: Robotic Bar-stools Enhance Intimacy in Couples' Conflict

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ABSTRACT

Couples' ability to manage conflicts is fundamental in the discourse of romantic relationships. A supportive environment for conflict management highly depends on non-verbal communication cues. We explored the possibility to facilitate positive non-verbal cues by using autonomous furniture that do not alter the dyadic nature of the interaction. Couples discussed a subject of disagreement while seated on autonomous bar-stools. During the discussion, the stools rotated inward, directing partners toward one another. In the baseline condition, the bar-stools were inactive. Our findings showed that the inward rotation facilitated couples' non-verbal cues (e.g., eye-gaze, body-orientation), which highly contributed to their communication and decreased conflict intensity. The positive effect was also evident in the number of intimate touch occurrences. The bar-stools' movement led to a significant increase in hand-holding, kisses, and hugs. This work highlights the great potential of leveraging autonomous furniture for enhancing romantic interactions while preserving their intimate and emotional nature.

KEYWORDS

Human-Human-Robot Interaction; conflict; Romantic-relationships; Robotic Furniture.

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1 INTRODUCTION

Human social interactions are amongst the most important factors influencing general wellbeing [27]. While Human-human

interaction (HHI) can take many forms, there is a consensus that romantic relationships are amongst the most important ones. The quality of such relationships has a wide range of implications for people's mental and physical health [33]. Surprisingly, this type of HHI is underexplored in the context of Human-Human-Robot Interaction (HHRI).

A major factor that is known to impact the quality of romantic couples' relationships is their ability to manage conflicts [22]. Conflict experiences can drastically impact couples' emotional closeness and their general satisfaction with the relationship [12]. The conflict management by the couple can also profoundly impact their perception of the conflict, which determines whether the conflict would escalate or be resolved [5]. Healthy relationships typically involve partners' ability to engage in constructive and positive communication. However, managing conflicts successfully is challenging as it requires couples to create an environment that facilitates deep levels of communication, self-disclosure, and understanding the partner's perspective [28]. Such a facilitatory environment highly depends on the couples' communication cues [13, 16]. Previous work suggests that an estimated 65% of interpersonal communication is conveyed via non-verbal behaviors [1, 3, 23]. Central non-verbal cues include face orientation, body orientation, smiling, and eye contact [17]. These non-verbal cues can enhance or impair couples' communication, especially in sensitive interactions involving conflict [6, 10, 32]. Directing body orientation toward each other and maintaining eye contact were offered as effective methods for facilitating conflict management [15, 31]. However, when encountering conflict, these positive non-verbal behaviors are rarely part of the couples' natural communication due to the sensitive, emotional context [20]. We suggest that technology offers a unique opportunity for facilitating positive non-verbal behaviors. In this work, we explored whether technology can be leveraged to trigger viable human non-verbal cues that enhance communication between couples in the context of conflict.

Previous work has already suggested autonomous technologies for enhancing intimate human interactions. These studies typically used robots as additional participants that mediate the interaction [11, 30]. Robots were designed to have an active role in HHI, such as a couples' counselor that speaks with the couple in order to prompt positive communication [30]. Other studies suggested a

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Figure 1: Left: A pair of robotic bar-stools. Center: The motors, gear, and microcontroller hidden within the inner side of the bar-stools. Used with permission [26]. Right: A couple in a relationship.

more peripheral approach where the robot performed simple gestures designed to raise awareness of aggressive communication (e.g. shaking when the volume of the discussion between partners is rising) [11]. While these studies indicated that robots can enhance romantic HHI, they were not leveraged to trigger positive non-verbal cues. In addition, these robotic solutions alter the dyadic nature of the couples' interaction as they add an additional entity (the robot) to an intimate and sensitive interaction.

A somewhat different approach that does not involve adding another (robotic) participant to the interaction was tested in non-intimate HHI [26, 29]. This approach leverages existing objects that are inherent to the interaction's context. The technology is embedded in the objects in a transparent way that doesn't alter the nature of the HHI [26]. For example, autonomous furniture were suggested as a method for enhancing interactions by moving participants into a spatial position that indicates a willingness for interaction. In the study, two strangers sitting on bar-stools were turned to face one another by a small rotation of the stools' seat. The rotation manipulated their body orientation towards one another, triggering a non-verbal communication that resulted in a natural positive opening encounter [26]. The positive impact on the strangers' interaction suggests that there is great potential in leveraging such autonomous furniture in other types of HHI.

In this study, we adopted the autonomous furniture approach. We explored the possibility of impacting a sensitive couple's interaction by embedding autonomous technology in the interaction's environment. Instead of adding a third (robotic) party to the interaction we used robotic bar-stools designed to rotate couples sitting on them while discussing a subject of disagreement (see Figure 1, used with permission [26]). The rotation of the bar-stools changed the couples' body orientation, triggering non-verbal cues that can potentially enhance communication and moderate conflicts (i.e., body position and eye-gaze). We evaluated the effect of the robotic bar-stools on the couple's interaction and compared it to a *Baseline* group where the stools were inactive.

2 RELATED WORK

Relevant work includes the impact of robots on the interaction between couples and the use of autonomous furniture for changing participants' body orientation in HHI:

2.1 Incorporating Robots into Couples Interactions

Several studies tested the impact of robots on the interaction between humans [7, 8, 25]. In the context of couple interaction, Utami and Bickmore (2019) developed a humanoid robot that was introduced as a counselor. The robot suggested exercises designed to facilitate the couple's interaction. Findings showed that participants engaged in the exercises led by the robot and that the robot was effective at increasing their sense of intimacy [30]. Another study by Hoffman et al. (2015) incorporated a peripheral non-humanoid robot into a couple's sensitive interaction that involved a conflict. The robot was located between couples during a discussion concerning a subject of disagreement and signaled when the conversation became aggressive and loud. Their findings showed that the robot's responsiveness increased participants' attention to the robot without altering the conversation's content [11].

We extend these studies by exploring an autonomous technology that does not present an additional entity to couples' sensitive interactions. We evaluated the possibility to leverage autonomous furniture that alter participants' body orientation and prompt non-verbal cues associated with intimacy.

2.2 Leveraging Autonomous Furniture for Enhancing Non-romantic HHI

Previous research has explored the potential use of autonomous furniture for encouraging and enhancing human-human interactions. For example, Takeuchi and You (2014) presented a design concept of facilitating social interactions in an urban setting using multiple autonomous stools. When a person sits on one of the stools, the orientation of all the stools changes, orienting people towards one another. They hypothesized that this change in body orientation would be sufficient to initiate a spontaneous social interaction [29]. This assumption was empirically tested by Sadka et al. (2021), who designed two robotic bar-stools that could turn the body orientation of the people sitting on them. In their study, they used the stools in the context of opening encounters between strangers. Their findings showed that the minimal movement toward one another facilitated positive social interactions and increased interpersonal communication [26].



Figure 2: Left: A couple before the movement of the stools. Right: A couple after the movement of the stools.

These studies demonstrate that embedding autonomous technology in the environment of HHI can serve as a tool for triggering non-verbal social cues known to facilitate social interactions. We extend this approach by applying autonomous furniture to romantic relationships and testing its impact in the sensitive context of conflict. Couples were invited to discuss a topic of disagreement with the researcher while seated on robotic bar-stools (see Figure 2, Left). The researcher left the room at a certain point in the discussion, and shortly after, the stools rotated, changing the participants' body orientation towards each other. The short and simple stool rotation was designed to trigger non-verbal behavior (body orientation and eye gaze) that can facilitate direct communication and enhance conflict management. We compared participants' experience, level of conflict, and behavior in two conditions: (1) 'Orientation towards one another' (*Inward* condition) and (2) 'No-movement' (*Baseline* condition).

3 METHOD

In order to gain preliminary insights into the possibility of using autonomous furniture to enhance couples' communication in the context of conflict, we conducted a study with 10 couples who were in a long-term relationship. The study was approved by the ethics committee of the research institute.

3.1 The Robotic Bar-stools

The pair of robotic bar-stools were designed following the approach of ubiquitous autonomous furniture. The bar-stools did not have backrests in order to allow participants to resist the movement and return to their original position. The stools' rotation was performed by a stepper motor that was connected via a chain to a gear that was attached to the stool's sitting area. The bar-stools could perform a synchronized rotation where the right bar-stool rotates counterclockwise and the left bar-stool rotates clockwise (45°). The bar-stools' movement was triggered through a wireless connection using a Wizard-of-Oz (WoZ) technique, a common method in HCI [18, 24].

3.2 Participants

We recruited a total of 10 couples (20 participants, 11 males, 9 females, 9 couples were heterosexual and one couple was homosexual, mean age = 23.9, SD = 3.1) that have been in a relationship for over

a year (mean years of relationship = 3.2, SD = 1.7). Student participants received extra course credits for participation in the study. Non-student participants received a voucher of 20\$. All couples signed a consent form.

3.3 Experimental Design

Two conditions were tested in a between-participants experimental design. In the *Inward* condition, the bar-stools changed participants' body orientation from a position facing the researcher towards each other (see Figure 2). In the *Baseline* condition, the bar-stools were inactive, and participants stayed in a position facing the researcher throughout the interaction. Participants were randomly assigned to one of the two conditions based on a pre-determined list.

3.4 Measures

Subjective conflict perception: Participants rated the intensity of conflict they experienced in the interaction before and after the short break (when the researcher left the room) on a 5-point Likert scale.

Semi-structured interview: To understand participants' experience, we conducted a post-experience semi-structured interview that allowed flexibility during data collection while remaining grounded in a particular framework [9]. Example questions: "Describe the experience"; "What did you think about the bar-stools?".

Intimate touch occurrences: The intimate physical touches that were coded were kisses, hugs, holding hands, and stroking. Since the intimate touch was sometimes mutual (e.g., kisses) and at other times non-mutual (e.g., stroking), we coded occurrences per couple and not per participant. We coded the number of intimate occurrences before and after the short break (when the researcher left the room).

3.5 Procedure

Upon arrival to the lab, couples were introduced to the researcher, signed an informed consent form, and filled in a demographic questionnaire. To design an interaction that involves conflict, we adapted Levenson and Gottmans (1983) [14] paradigm, where partners are asked to separately list subjects of disagreement. We asked participants to suggest topics they would feel comfortable discussing in the context of the experiment. Participants were then invited to the experimental room and were asked to sit on the bar-stools. The

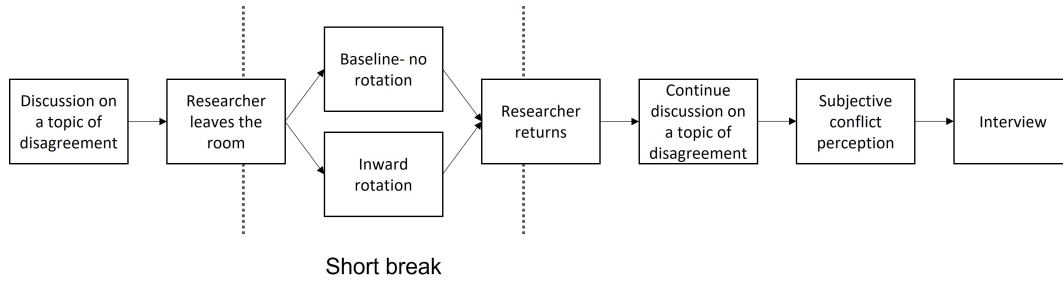


Figure 3: Experimental procedure

distance between the robotic bar-stools was set at 76 cm, as this is considered a comfortable distance in dyadic interactions [4]. A similar (non-robotic) bar-stool for the researcher was located in front of the couple at a distance of 110 cm from each of the robotic bar-stools. The researcher, together with the couple, chose one of the topics of disagreement and asked them to present it in an open discussion. Throughout the session, the researcher guided the interaction but did not take an active part in the couples' discussion. He responded in various ways, such as asking follow-up questions for clarification and asking for concrete examples. Despite being used in previous studies, we further validated this paradigm with an experienced couples' therapist (over 30 years of experience). Nine minutes after the discussion had started (a duration indicated in pilot studies as sufficient for a conflict experience to take place), a research assistant knocked on the door and asked for the researcher's help (which required leaving the room). In the *Inward* condition a research assistant activated the rotation of the robotic bar-stools forty seconds after the researcher left the room. In the *Baseline* condition, the bar-stools were inactive. One minute later, the researcher entered the room and asked the couple to continue the discussion. At the end of the session, the researcher handed the couple a tablet where they rated the conflict intensity. Then, a semi-structured interview was conducted. To end the experience in a positive atmosphere, the interview ended by asking the couple to share a positive mutual memory or experience. At the final stage of the experiment, the researcher debriefed the participants and verified that they left with an overall positive experience (see Figure 3).

4 FINDINGS

4.1 Quantitative Findings

The subjective level of conflict and the number of intimate touch occurrences were analyzed using a two-way-ANOVA.

Subjective conflict perception: Participants' ratings of the conflict intensity were analyzed using a two-way ANOVA evaluating the interaction between the stools conditions (*Inward*; *Baseline*) and time (before or after the short break where the researcher left the room). The analysis revealed a significant interaction, $F(1,28) = 3.41$, $p = 0.048$, $\eta_p^2 = 0.2$ (see Figure 4, Left). Scheffe's post-hoc analysis indicated that in the *Inward* condition, the conflict intensity significantly decreased after the break ($p=0.05$) while in the *Baseline* conflict intensity did not change ($p=0.55$). The main effects (stools conditions and conflict before and after the break) were not significant ($F<1$).

Intimate touch occurrences: We calculated the gap in the number of touch occurrences before and after the short break. Then, the gap was compared between the stools conditions (*Inward*; *Baseline*) using a Mann-Whitney analysis. The analysis revealed a significant effect, $U = 1.5$, $p = 0.027$ (see Figure 4, Right). In the *Inward* condition, the number of intimate touch occurrences significantly increased after the short break while in the *Baseline* it did not change.

4.2 Interview Findings

The semi-structured interview was analyzed using thematic coding [2], a methodology commonly used in HCI and HRI research. The analysis included five stages: (1) Interviews were transcribed, and half of the interviews were read several times by two coders to develop an overall understanding of the data before the coding process began; (2) Initial themes were identified, presented to a third researcher, and discussed in-depth until inconsistencies were resolved; (3) A list of mutually-agreed themes was defined; (4) The raters used these themes to analyze part of the data independently, verifying inter-rater reliability ($\kappa=87\%$); (5) The two coders analyzed the rest of the data. The interview analysis revealed three main themes: "Effect of the stools' movement on non-verbal cues", "Effect of the stools' movement on the couple's communication", and "Participants' sense of control".

4.3 Effect of the Stools' Movement on Non-verbal Cues

Participants talked about their own positive non-verbal cues that were triggered by the stools rotation. They mentioned eye gaze, proxemics, and body orientation. **Eye gaze:** "It's nice when I look at her. I'd rather look at her than you... (laughing)" (C.17,M); "I realized that it made us make eye contact" (C.14,F); **Closer proxemics:** "It brought me closer to him while we were talking" (C.8,F); **Body orientation:** "There's something about being face-to-face.." (C.8,M) ... "Yeah, being face-to-face is much better" (C.8,F); "I wanted it to move me even more towards his direction. It was nice" (C.6,F).

4.4 Effect of the Stools' Movement on the Couple's Communication

Participants explained how the non-verbal cues triggered by the stools affected their communication. They stated that the interaction became more intimate and promoted a feeling of closeness: "It made me focus on him. The movement made me look at him in a different way, we were closer" (C.8,F)... "Yes.. it became intimate"

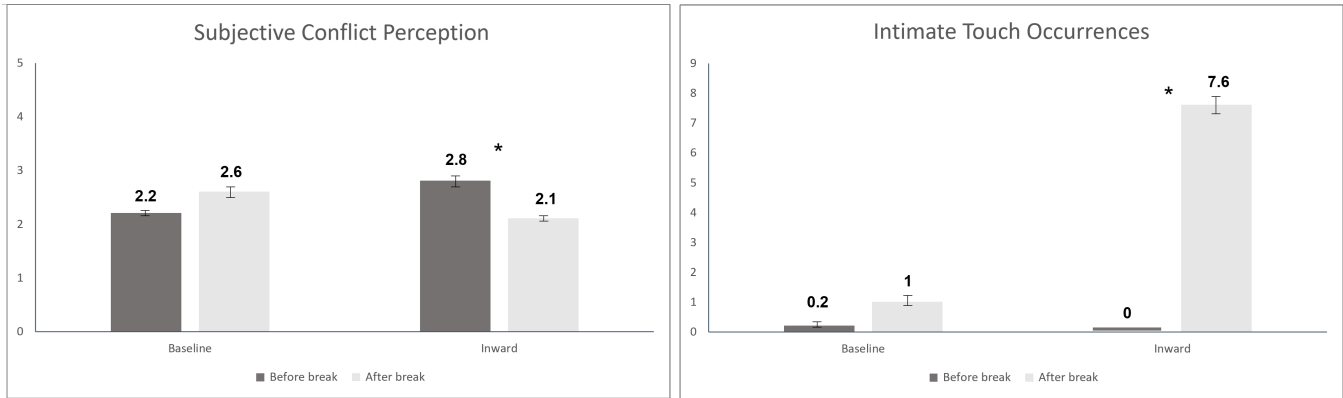


Figure 4: Left: Average ratings of subjective conflict perception. Right: Average of intimate touch occurrences.

(C.8,M). They further mentioned the stools signaled that the atmosphere should change: “Something physical (the stools) indicated that we should be in ‘good vibes’...” (C.14,F)...“It was really positive” (C.14,M). Most of them stated that after the stools moved the interaction was facilitated: “When he turned towards me it was much nicer” (C.6,F)...“Yeah, I really talked to her...together with her” (C.6,M); One couple mentioned that the movement made them kiss: “Facing each other felt natural” (C.17,F)...“Yeah, very natural” (C.17,M)...“So I gave him a kiss” (C.17,F). Another couple contextualized the stools’ movement within the conflict: “The movement made us take a moment and look at each other after discussing difficult topics” (C.4,M)...“To see how the other partner reacts” (C.4,F). One participant mentioned that the movement of the stools made him reflect on his behavior before the break: “I realized that throughout the conflict, I wasn’t looking at her. The stool positioned me in the place I should have been all along” (C.6,M). His partner then added: “I wanted to tell him to face me as we started talking. Then the stools rotated, they were there to help me” (C.6,F).

4.5 Participants’ Sense of Control

Participants discussed their lack of control over the movement of the stool. Most participants stated that the stools’ impact was positive despite the lack of control. Some implied that it was even essential: “People get upset and angry really quickly, something like this (the stools) that would force a different body orientation could make them snap out of it for a moment and think” (C.14,F). They stated that such a change in perspective is unlikely to be initiated by the couple. Thus, the uncontrolled movement is responsible for the positive effect: “During a conflict, you experience negative emotions. In those moments, you don’t stop and take a breath, saying to yourself that you love the other person. When the stools move it makes you come closer and calm down. You didn’t ask for it. It’s something physical in the room that wasn’t under your control. And it still worked” (C.14,F).

Three participants raised negative aspects of the uncontrolled stools’ movement: “When the stools rotated, it made me rotate with it. It made me feel weak since it forced me to be present in the interaction. I had no other choice” (C.8,M). They further explained that it is not suitable for everyone: “People have different emotional capacities, so you must consider that when forcing interactions” (C.8,F). They also explained that timing is important and should be controlled

by the couple: “Timing matters. The stool needs to rotate when you are ready to communicate with your partner” (C.17,F).

5 DISCUSSION

In this work, we explored whether autonomous furniture can enhance the interaction between couples in the context of conflict. We leveraged robotic bar-stools that were designed to change couples’ body orientation towards each other, triggering non-verbal cues known to facilitate deep communication and enhance conflict management. The ubiquitous technology, embedded in furniture, allowed to preserve the dyadic nature of a couple’s interaction.

Our findings showed that the rotation of the stools had a positive effect on the couples’ interaction. Participants were aware of the stools’ direct influence on their non-verbal behavior. They explicitly mentioned that the movement influenced their eye contact, proximity, and body orientation. Participants also explicitly stated that these cues influenced their conflict management by improving the atmosphere between them, making them look at each other in a different way, and increasing intimacy and closeness. For some, the movement of the stool even led to a general reflection on their behavior toward their partner. Such a positive impact on conflict management is known to enhance constructive communication and contribute to healthy romantic relationships [32]. The quantitative analysis further supported the positive effect. The rotation of the stools significantly increased intimate touch occurrences. Couples were more inclined to hug, kiss, and caress each other after the movement of the stools. In contrast, in the *Baseline* condition, intimate touch hardly occurred throughout the interaction. Finally, the subjective perception of the conflict intensity was also affected by the stools’ rotation. It significantly decreased after the stools rotated participants’ towards one another. This decrease was not evident in the *Baseline* condition, where the perception of the conflict intensity remained relatively constant.

Our preliminary findings also demonstrated the potential positive effect of embedding autonomous technologies in the natural HHI environment. Instead of incorporating an additional (robotic) entity to support couples’ interaction, the technology was embedded in bar-stools that increased partners’ attentiveness toward one another. Participants explicitly described how the technology directed their focus to each other and made them “see their partner”, “focus on their partner”, and “be together with their partner”. Thus,

the positive impact of the technology was mediated by their increased attention towards one another and not by the intervention of a third entity. Importantly, this ubiquitous implementation of the technology was well accepted by couples despite the intimate and sensitive context of a conflict. They also accepted the lack of control over the stools' movement and acknowledged the powerful impact of the manipulation of their body orientation. Some stated that such a positive impact would not be possible without the stools "forcing" the initiation of the non-verbal cues. However, it is important to consider that a few participants did not feel comfortable with the uncontrolled movement. Interestingly, the lack of a backrest that was designed to allow for resisting the movement was not perceived as sufficient for reinstating a sense of control. A possible way to overcome these adverse effects is to design the movement of the stool as a shared control interaction (see [34]) where the stools signal that they are about to take control and partners can accept or reject it.

In sum, we show the great potential in enhancing romantic relationships with autonomous furniture. When designed appropriately, technology can highly contribute to maintaining and improving one of the most important social relationships humans experience throughout their lives. Even emotional and sensitive interactions that involve conflict can dramatically benefit from well-designed technologies that focus on triggering positive communication cues and are ubiquitous to the environment. Specifically, the ability to preserve the dyadic nature of the interaction while supporting and enhancing it, is one of the greatest advantages of such autonomous technologies. At the same time, we show that the technology may threaten couples' sense of control and that further research should focus on adjusting the technology's level of autonomy.

6 LIMITATION AND FUTURE RESEARCH

This study has several limitations. First, in this study we explored one type of relationship (romantic), with presumably neurotypical and western participants, for whom intimate physical contact eases tension. Second, our findings do not provide information about the effect's persistence over time. Future studies should test autonomous furniture's impact on various types of relationships and over multiple conflict interactions. Third, the intensity of the conflict discussed in the interaction was mediated by the experiment's context and participants chose a subject of disagreement they felt comfortable sharing with the researcher. Further work should evaluate the stools' effects in a field study in couples' intimate environments. Another limitation concerns the interview. Participants' answers may have been affected by the "good subject effect" [19]. To minimize the effect, the interviewer followed a strict interview protocol and participants were reassured that anything said was helpful and valuable [21].

7 CONCLUSION

We demonstrate that autonomous furniture can be leveraged to facilitate couples' conflict management. By simply rotating partners towards each other, our robotic bar-stools triggered positive non-verbal cues while preserving the dyadic nature of the interaction. The enhanced communication between couples highlights the unique opportunity in such autonomous furniture.

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